

FRANKLIN, BENJAMIN (1706–1790)

Franklin is a key historical figure and one of the Founding Fathers of the US. He also had an avid interest in science and invention. Franklin began investigating electricity and, in 1756, by means of a kite and a key, demonstrated that lightning is electricity—a discovery that led to his invention of the lightning rod. Later inventions included bifocal glasses, a cast-iron stove, and a glass musical instrument called an armonica.

GOODE, SARAH E. (1855–1905)

This American inventor and entrepreneur is believed to be one of the first African American women to be granted a US patent. Not much is known about her life, but she was freed from slavery and moved to Chicago where she ran a furniture shop with her carpenter husband. In 1885, she was awarded the patent for her folding, hide-away bed. Her nifty invention could double as a bed or a folding writing desk thanks to a clever design that incorporated hinged sections that could be raised or lowered into place.

GUTENBERG, JOHANNES (c. 1395–c. 1468)

The German printer, whose “Forty-Two-Line” Bible (also known as the Gutenberg Bible) was the first book in Europe to be printed using movable type, perfected his printing press around 1439. Until Gutenberg invented his metal type, printing had been a laborious task using carved wooden blocks. Although his invention revolutionized printing, he died a poor man.

HANNAH, MARC (1956–)

Special effects in films such as *Jurassic Park* and *Beauty and the Beast* are partly thanks to African American electrical engineer and computer graphics designer Marc Hannah. He cofounded the company Silicon Graphics Inc. (SGI) in 1982 and helped pioneer the 3-D graphics technology used in many Hollywood blockbusters. More recently, Hannah has worked on multimedia plug-ins for listening to music on MP3 players and handheld video-game players.

HUYGENS, CHRISTIAAN (1629–1695)

The 17th-century Dutchman is regarded as one of the most important scientists and astronomers of all time. In the 1650s, he made improvements to telescopes that enabled him to see the rings and a moon orbiting Saturn. Huygens put forward groundbreaking theories about light waves and calculations for centrifugal force. He also excelled at invention, constructing the first pendulum clock (1657), prototypes for a small wristwatch, various telescopes, and a combustion engine that ran off gunpowder.

JONES, ELDORADO (1860–1932)

The American Eldorado Jones earned the nickname “the Iron Woman” because she invented a small portable iron, but it could also have been for her tough attitude on both work and men (she refused to employ them). She also invented a travel ironing board, an anti-damp salt shaker, and a collapsible hat rack. The airplane muffler, patented in 1923, was her most famous invention. It muffled the tremendous sound of an aircraft without causing loss of power.

KNIGHT, MARGARET (1838–1914)

Knight invented a machine to make flat-bottomed paper bags. The patent was awarded to Knight in 1871, making her the first woman in the US to be granted a patent, after a lengthy legal dispute with a man who tried to steal her idea. A prolific inventor, she also created a machine for cutting shoe soles, a safety device for cotton looms, a window frame, and various devices for engines.

KWOLEK, STEPHANIE (1923–2014)

This US chemist and polymer scientist nearly had a career in fashion, but opted for chemistry. Hired by the chemical company DuPont, in 1965 she discovered a new synthetic fiber called Kevlar. Strong yet lightweight, this fiber is used to make body armor for the police and military as well as airplanes, boats, and ropes.

LAKE, SIMON (1866–1945)

American marine engineer Simon Lake is remembered as the “Father of the Modern Submarine.” In 1894, he built his first submarine with a pressurized compartment, a vessel built for shallow waters that he named *Argonaut Jr.* Lake went on to create *Argonaut I*, which could be used at sea and moved on wheels along the ocean floor. In 1901, Lake produced a military sub called *Protector*, but it wasn’t until 1912 that he first had a submarine accepted by the US Navy.

